

# QuakeNet

## Earthquake Early Warning & Emergency Response Network

CSE421 — Computer Network

Section 07 | Group 703

*Project Report*

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## 1. Introduction

QuakeNet is a wide-area network designed to connect seven functional units of a national earthquake early-warning and emergency-response system: the National Emergency Operation Center (NEOC), Seismic Monitoring Station (SMS), Fire & Rescue Headquarters (FRH), Medical Emergency Center (MEC), Urban Search & Rescue Camp (USR), Communication Relay Center (CRC), and Mountain Observation Post (MOP).

The network was designed and built in Cisco Packet Tracer, and combines a redundant "warning loop" between the three most time-critical sites (NEOC, SMS, FRH) with a shared backbone switch, direct point-to-point links, VLSM-based IP addressing, RIPv2 with static-route redistribution, floating and recursive static backup routes, DHCP delivered through both router-based and server-based methods, and per-unit email/web/DNS services.

This report documents the design decisions, addressing scheme, full device configuration, and verification steps carried out to build and test the network.

## 2. Network Topology

Every unit is represented by one router, with its own LAN switch, two PCs, one network printer, and one server. Four units — NEOC, FRH, MEC, and CRC — additionally connect to a shared Central Switch, forming a backbone. Three further link types create the rest of the topology:

- Warning loop: three direct point-to-point links forming a triangle between NEOC, SMS, and FRH, layered on top of the shared switch connections NEOC and FRH already have — deliberate redundancy for the most time-critical alerting path.
- Direct fibre link: a dedicated point-to-point connection between NEOC and MEC, in addition to both already sharing the Central Switch.
- Stub links: USR connects only to MEC; MOP connects only to CRC.

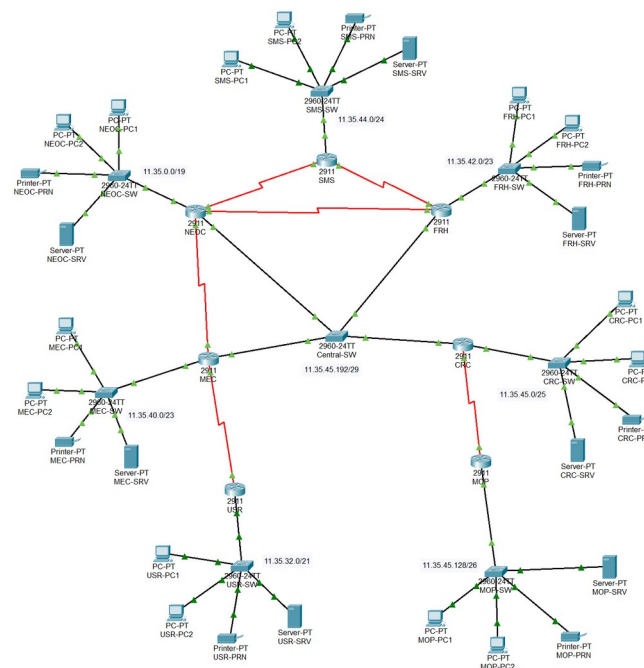


Figure 1 — QuakeNet network topology as built in Cisco Packet Tracer, with every device and subnet label as they appear on the live canvas.

Assumption: the spec describes the NEOC–MEC link as "direct fibre." It is implemented as a serial point-to-point link for consistency with the router hardware used across the rest of the design (Cisco 2911 routers with HWIC-2T serial modules) — no requirement is changed by this choice, only the simulated Layer-1 medium.

### 3. IP Addressing Scheme

#### 3.1 Base network and key calculations

Base network address, derived from the last group member's ID (23101135): last four digits 1135 → split into 11 and 35 → 11.35.0.0/16.

Custom Administrative Distance, derived from the first group member's ID (22299066): digit sum  $2+2+2+9+9+0+6+6 = 36$ ,  $\times$  group size 4 = 144. This value is used for every floating static route in Section 7.

#### 3.2 VLSM allocation

Subnets were allocated from 11.35.0.0/16 largest-requirement-first, the standard VLSM procedure: each block is sized to the smallest power of two that satisfies its stated host requirement, and the next block begins immediately after the previous one ends.

| Segment        | Hosts req. | Prefix | Network      | Usable range                | Broadcast    |
|----------------|------------|--------|--------------|-----------------------------|--------------|
| NEOC-LAN       | 5000       | /19    | 11.35.0.0    | 11.35.0.1 – 11.35.31.254    | 11.35.31.255 |
| USR-LAN        | 1040       | /21    | 11.35.32.0   | 11.35.32.1 – 11.35.39.254   | 11.35.39.255 |
| MEC-LAN        | 500        | /23    | 11.35.40.0   | 11.35.40.1 – 11.35.41.254   | 11.35.41.255 |
| FRH-LAN        | 320        | /23    | 11.35.42.0   | 11.35.42.1 – 11.35.43.254   | 11.35.43.255 |
| SMS-LAN        | 180        | /24    | 11.35.44.0   | 11.35.44.1 – 11.35.44.254   | 11.35.44.255 |
| CRC-LAN        | 100        | /25    | 11.35.45.0   | 11.35.45.1 – 11.35.45.126   | 11.35.45.127 |
| MOP-LAN        | 50         | /26    | 11.35.45.128 | 11.35.45.129 – 11.35.45.190 | 11.35.45.191 |
| Central switch | 4          | /29    | 11.35.45.192 | 11.35.45.193 – 11.35.45.198 | 11.35.45.199 |
| P2P NEOC–SMS   | 2          | /30    | 11.35.45.200 | 11.35.45.201 – 11.35.45.202 | 11.35.45.203 |
| P2P SMS–FRH    | 2          | /30    | 11.35.45.204 | 11.35.45.205 – 11.35.45.206 | 11.35.45.207 |
| P2P FRH–NEOC   | 2          | /30    | 11.35.45.208 | 11.35.45.209 – 11.35.45.210 | 11.35.45.211 |
| P2P NEOC–MEC   | 2          | /30    | 11.35.45.212 | 11.35.45.213 – 11.35.45.214 | 11.35.45.215 |
| P2P USR–MEC    | 2          | /30    | 11.35.45.216 | 11.35.45.217 – 11.35.45.218 | 11.35.45.219 |
| P2P CRC–MOP    | 2          | /30    | 11.35.45.220 | 11.35.45.221 – 11.35.45.222 | 11.35.45.223 |

Table 1 — Full VLSM subnet allocation.

The tree below shows the same allocation as a branching diagram: starting from the full 11.35.0.0/16 block, each split halves the address space (borrowing one more host bit), continuing down whichever half still contains further requirements until every subnet in Table 1 is reached. Branches that reach a size no longer needed by any requirement are left unused rather than split further.

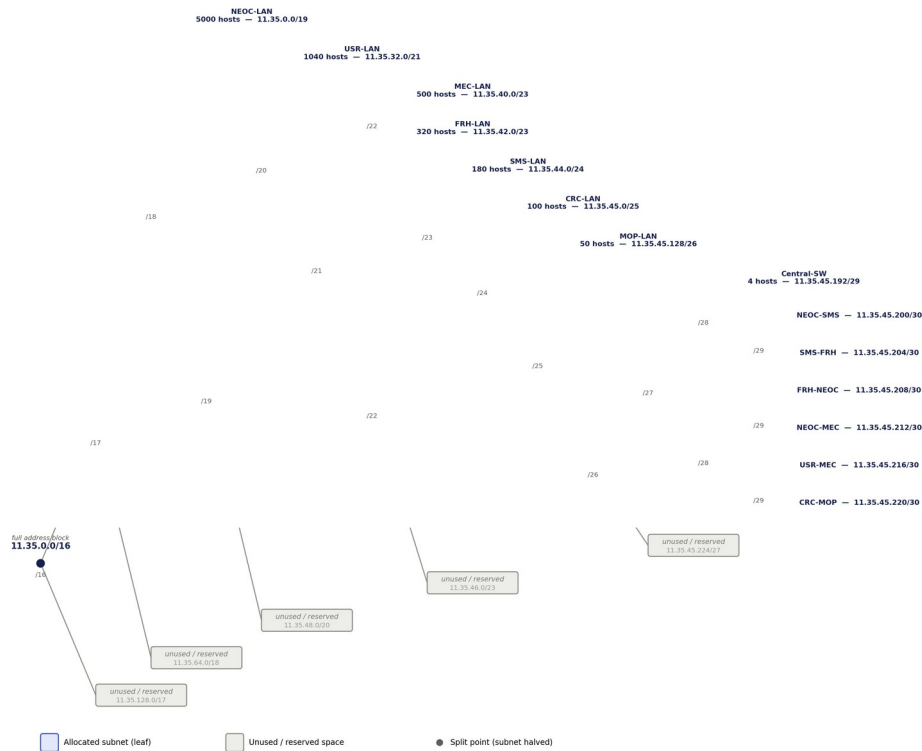


Figure 2 — VLSM tree: the 11.35.0.0/16 block subdivided down to every allocated subnet.

### 3.3 Per-site static addressing convention

Within every LAN subnet: the first usable address is the router's gateway interface, the second is that unit's server, the third is its printer, and the remainder of the range is the DHCP pool for its two PCs.

| Device   | IP           | Mask            | Gateway      | DNS       |
|----------|--------------|-----------------|--------------|-----------|
| NEOC-SRV | 11.35.0.2    | 255.255.224.0   | 11.35.0.1    | 11.35.0.2 |
| NEOC-PRN | 11.35.0.3    | 255.255.224.0   | 11.35.0.1    | 11.35.0.2 |
| SMS-SRV  | 11.35.44.2   | 255.255.255.0   | 11.35.44.1   | 11.35.0.2 |
| SMS-PRN  | 11.35.44.3   | 255.255.255.0   | 11.35.44.1   | 11.35.0.2 |
| FRH-SRV  | 11.35.42.2   | 255.255.254.0   | 11.35.42.1   | 11.35.0.2 |
| FRH-PRN  | 11.35.42.3   | 255.255.254.0   | 11.35.42.1   | 11.35.0.2 |
| MEC-SRV  | 11.35.40.2   | 255.255.254.0   | 11.35.40.1   | 11.35.0.2 |
| MEC-PRN  | 11.35.40.3   | 255.255.254.0   | 11.35.40.1   | 11.35.0.2 |
| USR-SRV  | 11.35.32.2   | 255.255.248.0   | 11.35.32.1   | 11.35.0.2 |
| USR-PRN  | 11.35.32.3   | 255.255.248.0   | 11.35.32.1   | 11.35.0.2 |
| CRC-SRV  | 11.35.45.2   | 255.255.255.128 | 11.35.45.1   | 11.35.0.2 |
| CRC-PRN  | 11.35.45.3   | 255.255.255.128 | 11.35.45.1   | 11.35.0.2 |
| MOP-SRV  | 11.35.45.130 | 255.255.255.192 | 11.35.45.129 | 11.35.0.2 |
| MOP-PRN  | 11.35.45.131 | 255.255.255.192 | 11.35.45.129 | 11.35.0.2 |

Table 2 — Static addresses for every server and printer.

## 4. Device Inventory

Every router is a Cisco 2911. NEOC carries two HWIC-2T serial modules (three serial links); every other router carries one (one or two serial links). All switches are 2960-series. 43 devices in total: 7 routers, 8 switches (7 LAN + 1 central), 14 PCs, 7 printers, 7 servers.

| Site | Router interfaces used               | LAN switch | End devices               |
|------|--------------------------------------|------------|---------------------------|
| NEOC | Gi0/0, Gi0/1, S0/1/0, S0/1/1, S0/2/0 | NEOC-SW    | 2 PC, 1 printer, 1 server |
| SMS  | Gi0/0, S0/1/0, S0/1/1                | SMS-SW     | 2 PC, 1 printer, 1 server |
| FRH  | Gi0/0, Gi0/1, S0/1/0, S0/1/1         | FRH-SW     | 2 PC, 1 printer, 1 server |
| MEC  | Gi0/0, Gi0/1, S0/1/0, S0/1/1         | MEC-SW     | 2 PC, 1 printer, 1 server |
| USR  | Gi0/0, S0/1/0                        | USR-SW     | 2 PC, 1 printer, 1 server |
| CRC  | Gi0/0, Gi0/1, S0/1/0                 | CRC-SW     | 2 PC, 1 printer, 1 server |
| MOP  | Gi0/0, S0/1/0                        | MOP-SW     | 2 PC, 1 printer, 1 server |

Table 3 — Device inventory by site.

## 5. Router Configurations

Full final running-configuration for each router (interfaces, DHCP, static routes, and RIP where applicable), reflecting the cumulative state after every configuration phase.

### 5.1 NEOC

```

hostname NEOC
!
interface GigabitEthernet0/0
 ip address 11.35.0.1 255.255.224.0
 no shutdown
!
interface GigabitEthernet0/1
 ip address 11.35.45.193 255.255.255.248
 no shutdown
!
interface Serial0/1/0
 ip address 11.35.45.201 255.255.255.252
 clock rate 64000
 no shutdown
!
interface Serial0/1/1
 ip address 11.35.45.210 255.255.255.252
 clock rate 64000
 no shutdown
!
interface Serial0/2/0
 ip address 11.35.45.213 255.255.255.252
 clock rate 64000
 no shutdown
!
ip dhcp excluded-address 11.35.0.1 11.35.0.3
ip dhcp excluded-address 11.35.44.1 11.35.44.3
ip dhcp excluded-address 11.35.42.1 11.35.42.3
!
ip dhcp pool NEOC-P00L
 network 11.35.0.0 255.255.224.0
 default-router 11.35.0.1
 dns-server 11.35.0.2

```

```

!
ip dhcp pool SMS-P00L
 network 11.35.44.0 255.255.255.0
 default-router 11.35.44.1
 dns-server 11.35.0.2
!
ip dhcp pool FRH-P00L
 network 11.35.42.0 255.255.254.0
 default-router 11.35.42.1
 dns-server 11.35.0.2
!
ip route 11.35.40.0 255.255.254.0 11.35.45.214
ip route 11.35.32.0 255.255.248.0 11.35.45.214
ip route 11.35.45.0 255.255.255.128 11.35.45.196
ip route 11.35.45.128 255.255.255.192 11.35.45.196
!
router rip
 version 2
 network 11.0.0.0
 passive-interface GigabitEthernet0/0
 passive-interface GigabitEthernet0/1
 passive-interface Serial0/2/0
 redistribute static
 default-metric 1
 no auto-summary

```

## 5.2 SMS

```

hostname SMS
!
interface GigabitEthernet0/0
 ip address 11.35.44.1 255.255.255.0
 ip helper-address 11.35.45.201
 no shutdown
!
interface Serial0/1/0
 ip address 11.35.45.202 255.255.255.252
 no shutdown
!
interface Serial0/1/1
 ip address 11.35.45.205 255.255.255.252
 clock rate 64000
 no shutdown
!
router rip
 version 2
 network 11.0.0.0
 passive-interface GigabitEthernet0/0
 no auto-summary

```

## 5.3 FRH

```

hostname FRH
!
interface GigabitEthernet0/0
 ip address 11.35.42.1 255.255.254.0
 ip helper-address 11.35.45.210
 no shutdown
!

```

```

interface GigabitEthernet0/1
 ip address 11.35.45.194 255.255.255.248
 no shutdown
!
interface Serial0/1/0
 ip address 11.35.45.206 255.255.255.252
 no shutdown
!
interface Serial0/1/1
 ip address 11.35.45.209 255.255.255.252
 no shutdown
!
router rip
 version 2
 network 11.0.0.0
 passive-interface GigabitEthernet0/0
 passive-interface GigabitEthernet0/1
 no auto-summary

```

## 5.4 MEC

```

hostname MEC
!
interface GigabitEthernet0/0
 ip address 11.35.40.1 255.255.254.0
 no shutdown
!
interface GigabitEthernet0/1
 ip address 11.35.45.195 255.255.255.248
 no shutdown
!
interface Serial0/1/0
 ip address 11.35.45.214 255.255.255.252
 no shutdown
!
interface Serial0/1/1
 ip address 11.35.45.218 255.255.255.252
 clock rate 64000
 no shutdown
!
ip route 11.35.0.0 255.255.224.0 11.35.45.213
ip route 11.35.44.0 255.255.255.0 11.35.45.213
ip route 11.35.42.0 255.255.254.0 11.35.45.213
ip route 11.35.45.0 255.255.255.128 11.35.45.196
ip route 11.35.45.128 255.255.255.192 11.35.45.196
ip route 11.35.32.0 255.255.248.0 11.35.45.217
ip route 11.35.0.0 255.255.224.0 11.35.45.196 144
ip route 11.35.44.0 255.255.255.0 11.35.45.196 144
ip route 11.35.42.0 255.255.254.0 11.35.45.196 144

```

## 5.5 USR

```

hostname USR
!
interface GigabitEthernet0/0
 ip address 11.35.32.1 255.255.248.0
 ip helper-address 11.35.40.2
 no shutdown
!

```



```

interface Serial0/1/0
 ip address 11.35.45.217 255.255.255.252
 no shutdown
!
ip route 11.35.0.0 255.255.224.0 Serial0/1/0
ip route 11.35.44.0 255.255.255.0 Serial0/1/0
ip route 11.35.42.0 255.255.254.0 Serial0/1/0
ip route 11.35.40.0 255.255.254.0 Serial0/1/0
ip route 11.35.45.0 255.255.255.128 Serial0/1/0
ip route 11.35.45.128 255.255.255.192 Serial0/1/0

```

## 5.6 CRC

```

hostname CRC
!
interface GigabitEthernet0/0
 ip address 11.35.45.1 255.255.255.128
 no shutdown
!
interface GigabitEthernet0/1
 ip address 11.35.45.196 255.255.255.248
 no shutdown
!
interface Serial0/1/0
 ip address 11.35.45.221 255.255.255.252
 clock rate 64000
 no shutdown
!
ip route 11.35.0.0 255.255.224.0 11.35.45.193
ip route 11.35.44.0 255.255.255.0 11.35.45.193
ip route 11.35.42.0 255.255.254.0 11.35.45.193
ip route 11.35.40.0 255.255.254.0 11.35.45.195
ip route 11.35.32.0 255.255.248.0 11.35.45.195
ip route 11.35.45.128 255.255.255.192 11.35.45.222
ip route 11.35.0.0 255.255.224.0 11.35.45.194 144
ip route 11.35.44.0 255.255.255.0 11.35.45.194 144
ip route 11.35.42.0 255.255.254.0 11.35.45.194 144

```

## 5.7 MOP

```

hostname MOP
!
interface GigabitEthernet0/0
 ip address 11.35.45.129 255.255.255.192
 ip helper-address 11.35.45.2
 no shutdown
!
interface Serial0/1/0
 ip address 11.35.45.222 255.255.255.252
 no shutdown
!
ip route 11.35.0.0 255.255.224.0 Serial0/1/0
ip route 11.35.44.0 255.255.255.0 Serial0/1/0
ip route 11.35.42.0 255.255.254.0 Serial0/1/0
ip route 11.35.40.0 255.255.254.0 Serial0/1/0
ip route 11.35.32.0 255.255.248.0 Serial0/1/0
ip route 11.35.45.0 255.255.255.128 Serial0/1/0

```

## 6. DHCP Configuration

DHCP is delivered two ways in this design, deliberately: router-based (native IOS) for the RIP-domain sites, and server-based (Packet Tracer Server service) elsewhere.

| Server     | Pool      | Gateway      | DNS       | Start IP     | Mask | Max users |
|------------|-----------|--------------|-----------|--------------|------|-----------|
| NEOC (IOS) | NEOC-POOL | 11.35.0.1    | 11.35.0.2 | 11.35.0.4    | /19  | —         |
| NEOC (IOS) | SMS-POOL  | 11.35.44.1   | 11.35.0.2 | 11.35.44.4   | /24  | —         |
| NEOC (IOS) | FRH-POOL  | 11.35.42.1   | 11.35.0.2 | 11.35.42.4   | /23  | —         |
| MEC-SRV    | MEC-POOL  | 11.35.40.1   | 11.35.0.2 | 11.35.40.4   | /23  | 500       |
| MEC-SRV    | USR-POOL  | 11.35.32.1   | 11.35.0.2 | 11.35.32.4   | /21  | 1040      |
| CRC-SRV    | CRC-POOL  | 11.35.45.1   | 11.35.0.2 | 11.35.45.4   | /25  | 100       |
| CRC-SRV    | MOP-POOL  | 11.35.45.129 | 11.35.0.2 | 11.35.45.132 | /26  | 50        |

Table 4 — All seven DHCP pools.

SMS, FRH, USR, and MOP relay client broadcasts via ip helper-address (see Section 5) since they host no DHCP service of their own — SMS and FRH point at NEOC directly, USR at MEC-SRV, and MOP at CRC-SRV.

## 7. Routing Design

### 7.1 RIPv2 domain

NEOC, SMS, and FRH run RIPv2 over the warning-loop links. LAN-facing and non-RIP-domain interfaces are configured passive-interface so RIP neither forms unintended neighbor relationships (e.g. NEOC and FRH also share the Central Switch) nor advertises onto end-host segments — the connected subnets are still carried in RIP updates through the active interfaces regardless.

### 7.2 Redistribution

NEOC holds static routes to MEC-LAN, USR-LAN, CRC-LAN, and MOP-LAN (none of which are reachable via RIP), and redistributes them into RIPv2 with a seed metric of 1. This is how SMS and FRH — which never speak to MEC or CRC directly — learn that the rest of the network exists.

### 7.3 Floating and recursive static routes

MEC’s primary route to the NEOC/SMS/FRH networks uses the direct fibre link (next-hop 11.35.45.213); a floating, recursive backup route via CRC (next-hop 11.35.45.196) is installed at the custom Administrative Distance of 144, calculated in Section 3.1. It only appears in the routing table if the primary path is withdrawn.

CRC’s primary route to the same three networks uses NEOC directly across the Central Switch (11.35.45.193); its floating backup via FRH (11.35.45.194) also uses AD 144. Because both next-hops sit on the same shared switch segment, this backup demonstrates a real limitation of recursive floating statics: IOS considers the route resolved as long as some route covers the next-hop’s subnet, regardless of whether that specific host is alive, so the backup does not activate automatically the way MEC’s does. It was verified by temporarily withdrawing the primary route (see Section 8).

### 7.4 Static-only stubs

USR and MOP each have exactly one path out, so their routes to every other network use exit-interface syntax (Serial0/1/0) rather than a next-hop IP — a deliberate contrast with MEC and CRC’s recursive, next-hop-based routes. No default routes are used anywhere in the design.

## 8. Server Configuration

### 8.1 Email

All seven servers run the Email service. Each unit has two mailboxes (unit-prefixed, e.g. neoc1/neoc2), matched to that unit's two PCs.

| Server   | Domain        | Users        |
|----------|---------------|--------------|
| NEOC-SRV | mail.neoc.gov | neoc1, neoc2 |
| SMS-SRV  | mail.sms.gov  | sms1, sms2   |
| FRH-SRV  | mail.frh.gov  | frh1, frh2   |
| MEC-SRV  | mail.mec.gov  | mec1, mec2   |
| USR-SRV  | mail.usr.gov  | usr1, usr2   |
| CRC-SRV  | mail.crc.gov  | crc1, crc2   |
| MOP-SRV  | mail.mop.gov  | mop1, mop2   |

Table 5 — Email domains and accounts.

Cross-domain email between NEOC and MEC was verified with a round trip between neoc1 and mec2@mail.mec.gov (Section 9).

### 8.2 Web

NEOC-SRV and MEC-SRV run the HTTP service, serving [www.neoc.alert.gov](http://www.neoc.alert.gov) and [www.mec.alert.gov](http://www.mec.alert.gov) respectively, each with a short custom status page.

### 8.3 DNS

NEOC-SRV runs the network's only DNS service, authoritative for nine A-records: the seven mail\*.gov mailserver names and the two www\*.alert.gov web server names. Every device in the network — statically configured or DHCP-assigned — points to 11.35.0.2 as its DNS server.

## 9. Testing & Verification

The following checks were carried out to verify the design end-to-end.

### 9.1 Connectivity

- NEOC-PC1 → SMS-PC1 (warning loop / RIP)
- SMS-PC1 → FRH-PC1 (warning loop, other leg)
- NEOC-PC1 → CRC-PC1 (central switch)
- MEC-PC1 → USR-PC1 (stub link)
- CRC-PC1 → MOP-PC1 (stub link)
- SMS-PC1 → MOP-PC1 (full path: RIP → redistribution → static domain → stub)

```

C:\>ping 11.35.45.132

Pinging 11.35.45.132 with 32 bytes of data:

Reply from 11.35.45.132: bytes=32 time=21ms TTL=124
Reply from 11.35.45.132: bytes=32 time=63ms TTL=124
Reply from 11.35.45.132: bytes=32 time=74ms TTL=124
Reply from 11.35.45.132: bytes=32 time=2ms TTL=124

Ping statistics for 11.35.45.132:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 74ms, Average = 40ms

C:\>|

```

Figure 3 — Successful ping test confirming end-to-end connectivity across the network.

## 9.2 DHCP

All 14 PCs confirmed to hold a leased address from the correct pool (Desktop → IP Configuration).

## 9.3 Floating static route failover (MEC)

With NEOC's Serial0/2/0 administratively shut down, MEC's route to 11.35.0.0/19 was confirmed to switch from AD 1 (via 11.35.45.213) to AD 144 (via 11.35.45.196), and MEC-PC1 → NEOC-PC1 connectivity was confirmed to continue across the backup path. The primary link was then restored and the route confirmed to revert automatically.

```

MEC#show ip route 11.35.0.0
Routing entry for 11.35.0.0/19
Known via "static", distance 144, metric 0
  Routing Descriptor Blocks:
    * 11.35.45.196
      Route metric is 0, traffic share count is 1
MEC#|

```

Figure 4 — MEC's routing table showing the floating backup route active at administrative distance 144 once the primary link was withdrawn.

## 9.4 Floating static route (CRC) — backup verification

Because CRC's primary and backup next-hops share the Central Switch segment (Section 7.3), the backup route was verified by temporarily removing the primary ip route entries and confirming the AD-144 entries appeared alone in the routing table, then restoring the primary configuration.

## 9.5 Warning-loop resilience

With NEOC's Serial0/1/0 (to SMS) shut down, RIP reconverged and NEOC ↔ SMS traffic was confirmed to reroute via FRH.

## 9.6 Web and DNS verification

Both [www.neoc.alert.gov](http://www.neoc.alert.gov) and [www.mec.alert.gov](http://www.mec.alert.gov) were confirmed to load by name from a PC (FRH-PC1), verifying DNS resolution, HTTP, and routing together in a single test.

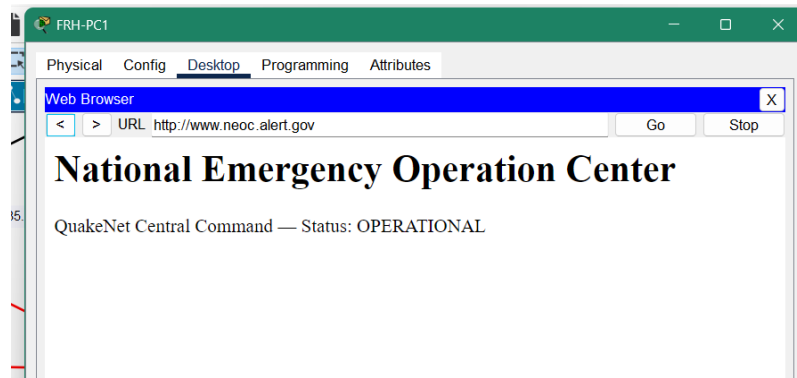


Figure 5 — [www.neoc.alert.gov](http://www.neoc.alert.gov) loading by name from FRH-PC1.

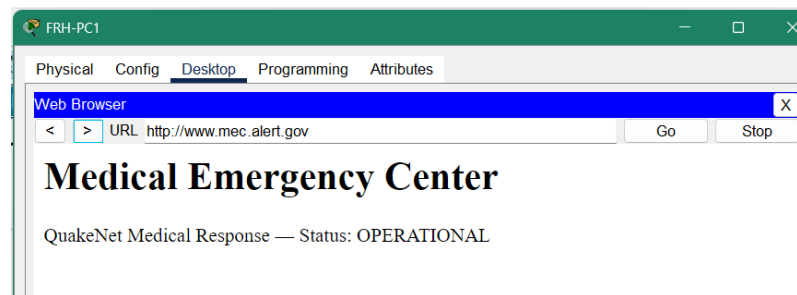


Figure 6 — [www.mec.alert.gov](http://www.mec.alert.gov) loading by name from FRH-PC1.

## 9.7 Email verification

Intra-unit email delivery was confirmed for each site, along with the NEOC ↔ MEC cross-domain round trip: mail sent from [neoc1@mail.neoc.gov](mailto:neoc1@mail.neoc.gov) was confirmed received in MEC-PC2's inbox ([mec2@mail.mec.gov](mailto:mec2@mail.mec.gov)), demonstrating that email routes correctly across site and domain boundaries.

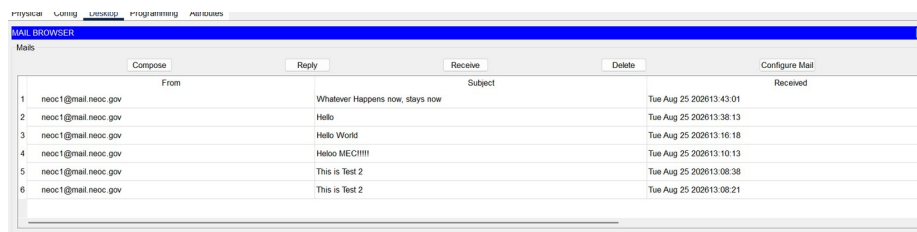


Figure 7 — MEC-PC2's inbox showing mail delivered from [neoc1@mail.neoc.gov](mailto:neoc1@mail.neoc.gov), confirming cross-domain email.

## 10. Assumptions

- The last group member for the base-network calculation and the first member for the Administrative Distance calculation were taken in the order the group listed its own members (Member 1 → Member 4).
- The NEOC–MEC "direct fibre" link is implemented as a serial point-to-point link rather than a literal fibre medium, for consistency with the router hardware used everywhere else.
- MEC and CRC's DHCP is server-based (Packet Tracer Server service) while NEOC's is router-based (IOS), to demonstrate both methods within one project.
- CRC's and MEC's floating backup routes toward the RIP domain use the same next-hop granularity as their primary routes (i.e. all three RIP-domain LANs share one primary and one backup next-hop) rather than routing FRH-LAN directly via FRH, treating the RIP domain as one logical destination reachable through either NEOC or FRH.
- A shared password (Passw0rd123) is used for all email accounts for simplicity, since credential security is not part of this assignment's scope.